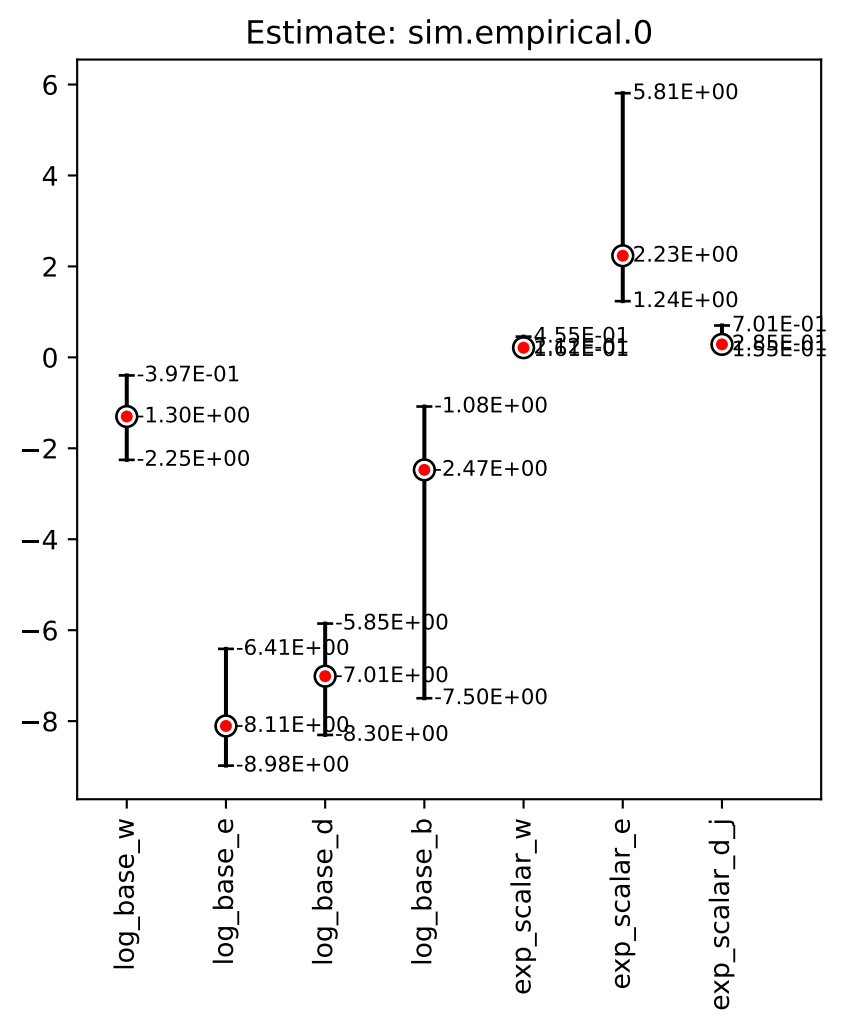
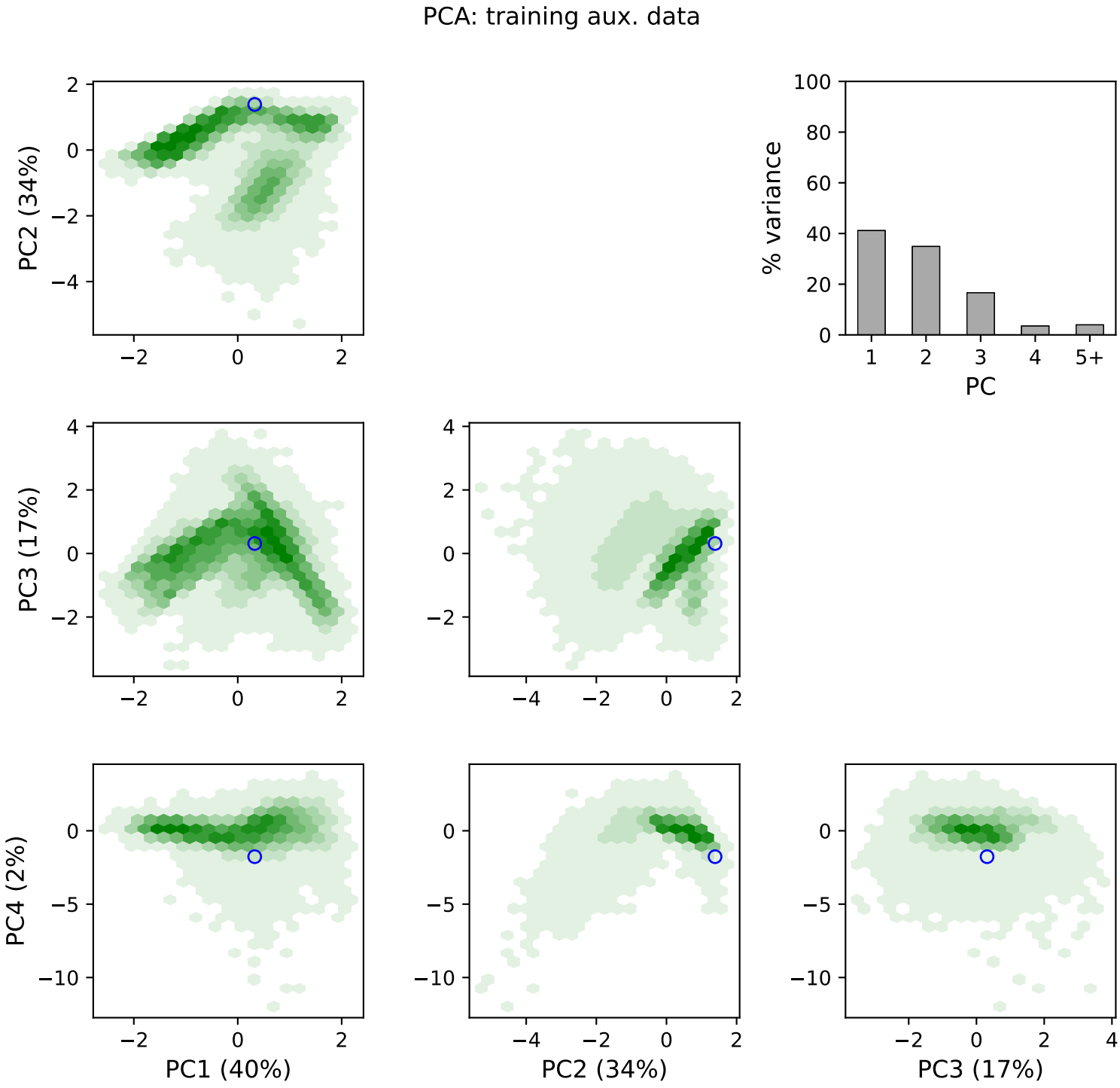
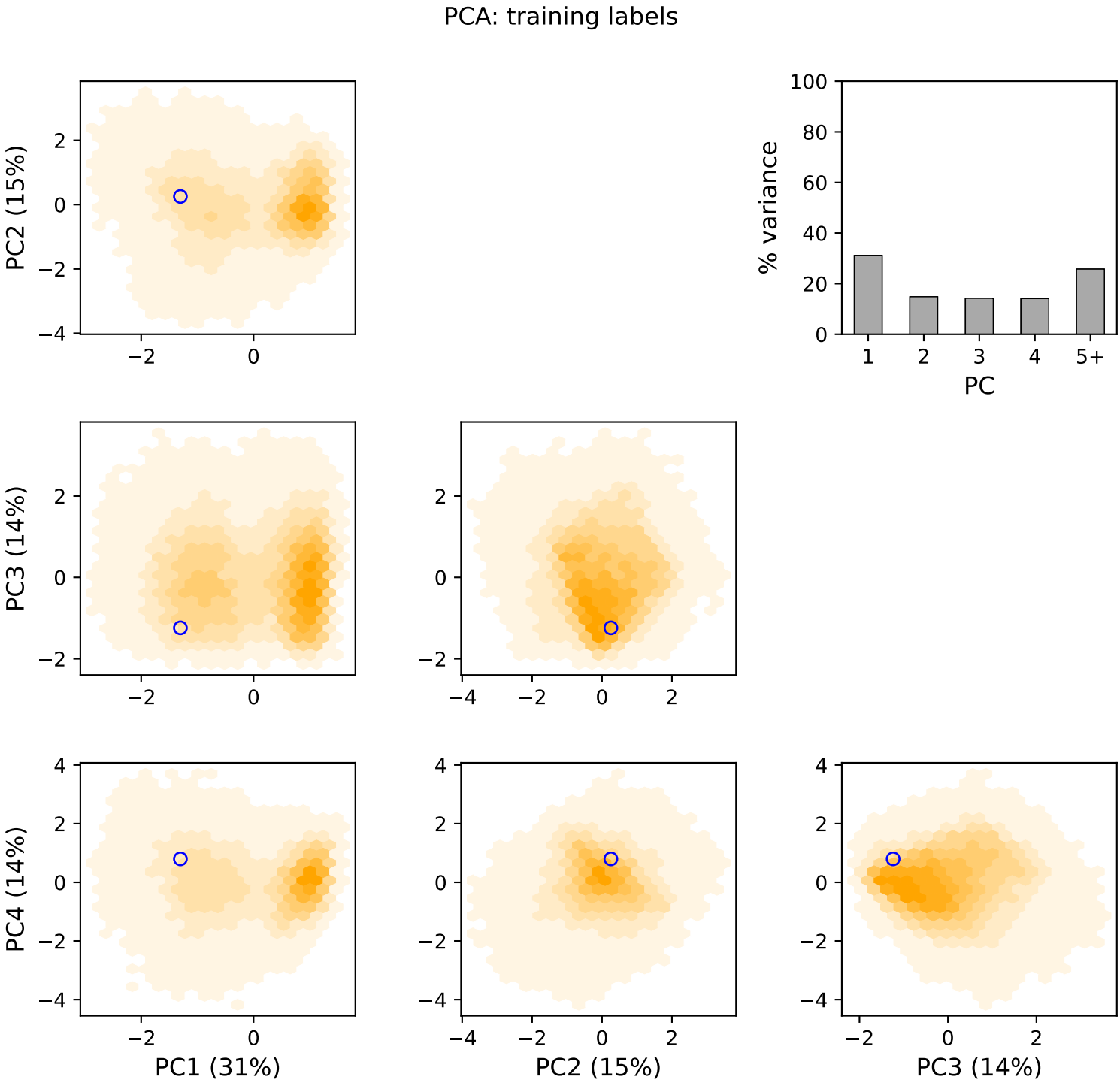
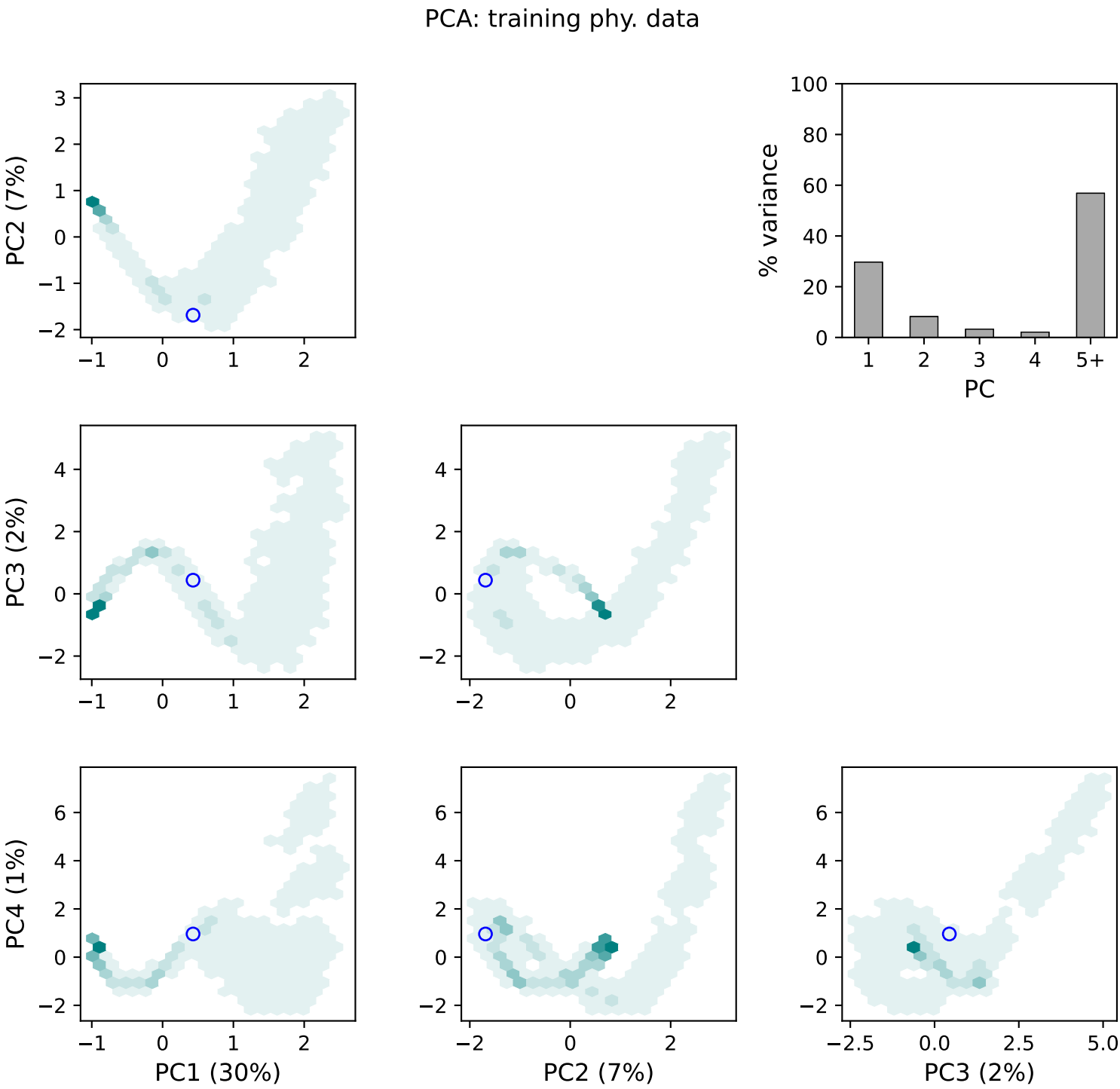
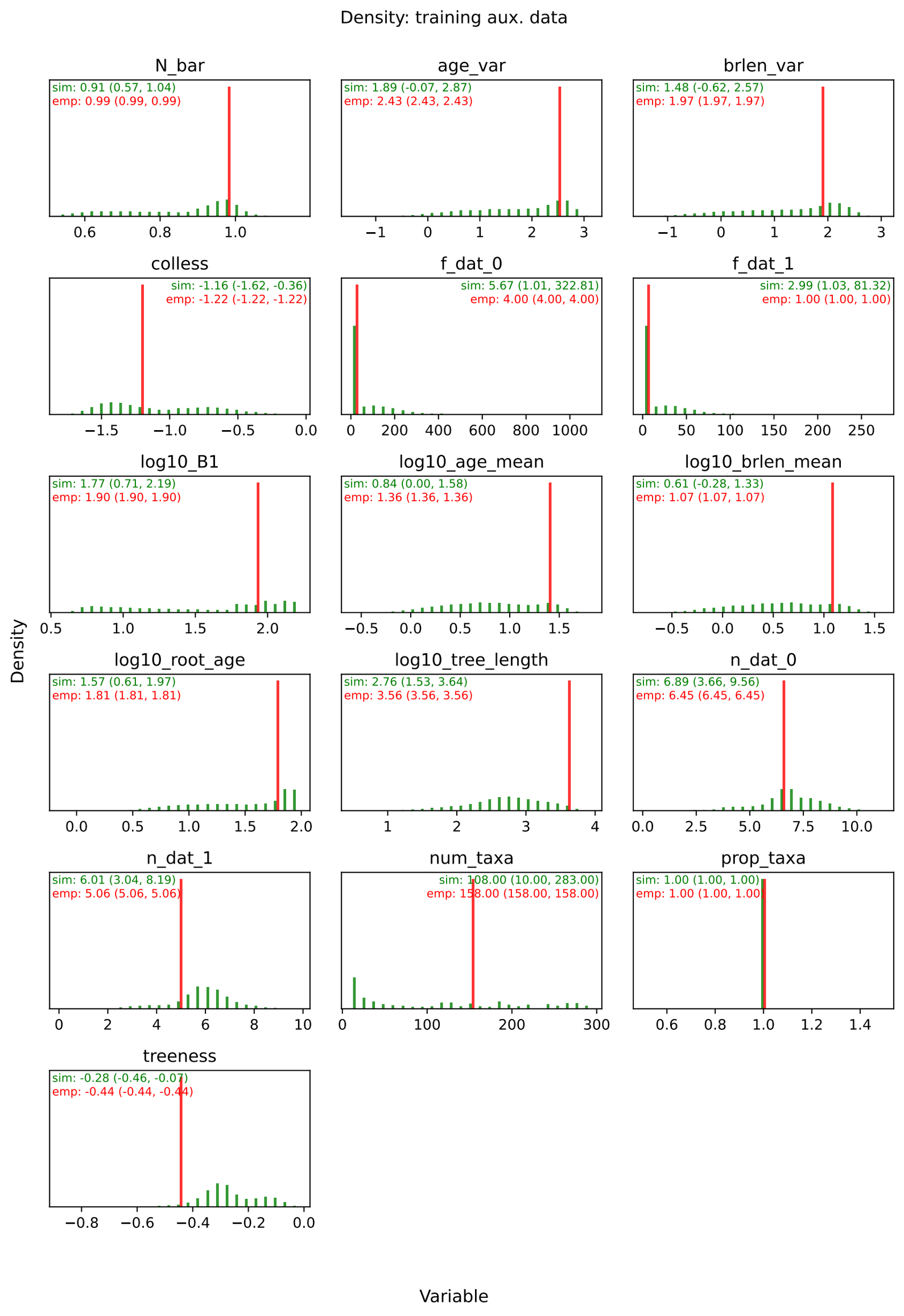
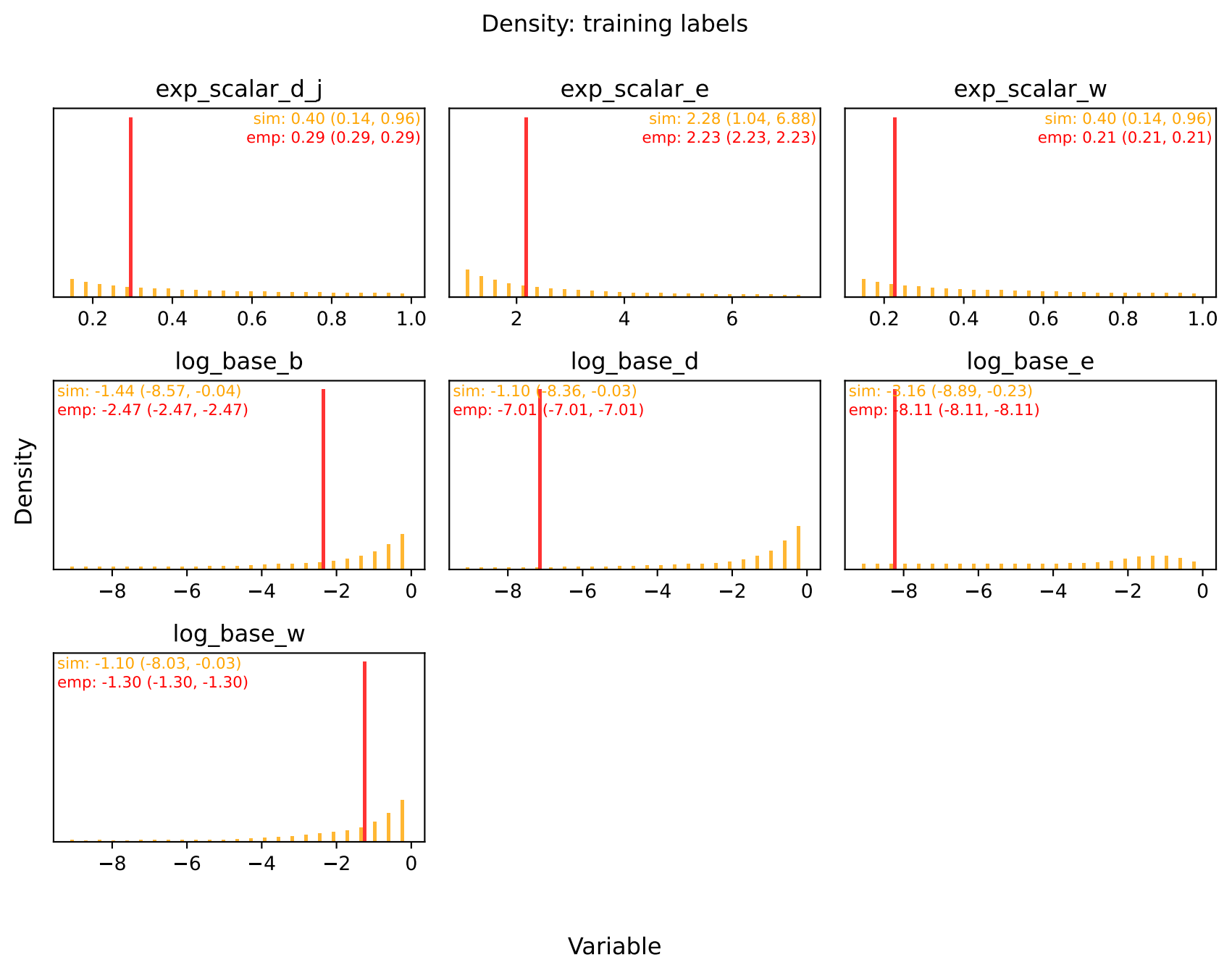

Estimate: sim.empirical.0
6
4
2
0
−2
−4
−6
−8
-3.97E-01
-1.30E+00
-2.25E+00
-6.41E+00
-8.11E+00
-8.98E+00
-5.85E+00
-7.01E+00
-8.30E+00
-1.08E+00
-2.47E+00
-7.50E+00
-4.55E-01
5.81E+00
2.23E+00
1.24E+00
7.01E-01
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w
exp_scalar_e
exp_scalar_d_j

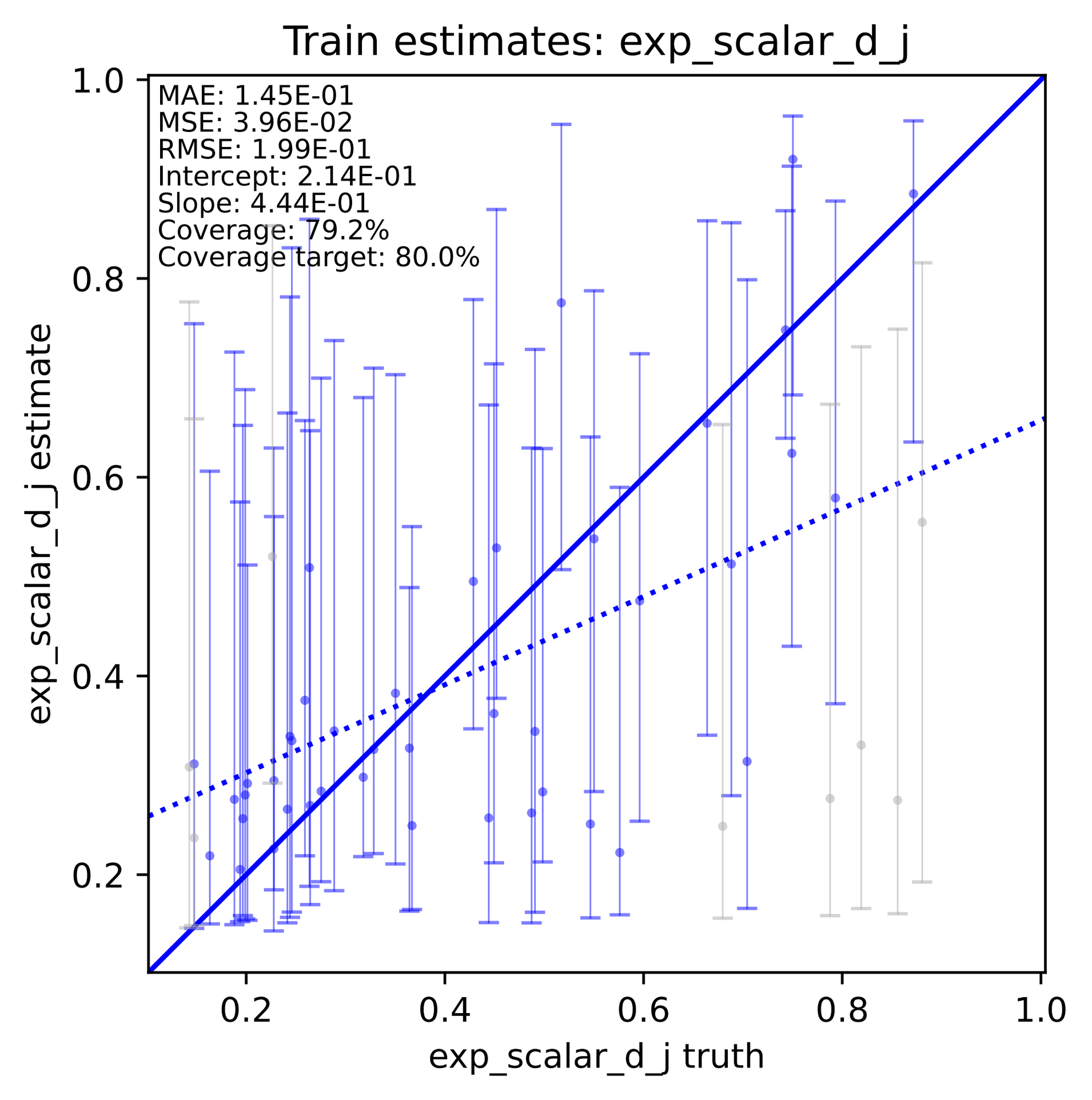

PCA: training aux. data
PC2 (34%)
% variance
PC
PC3 (17%)
PC4 (2%)
PC1 (40%)
PC2 (34%)
PC3 (17%)

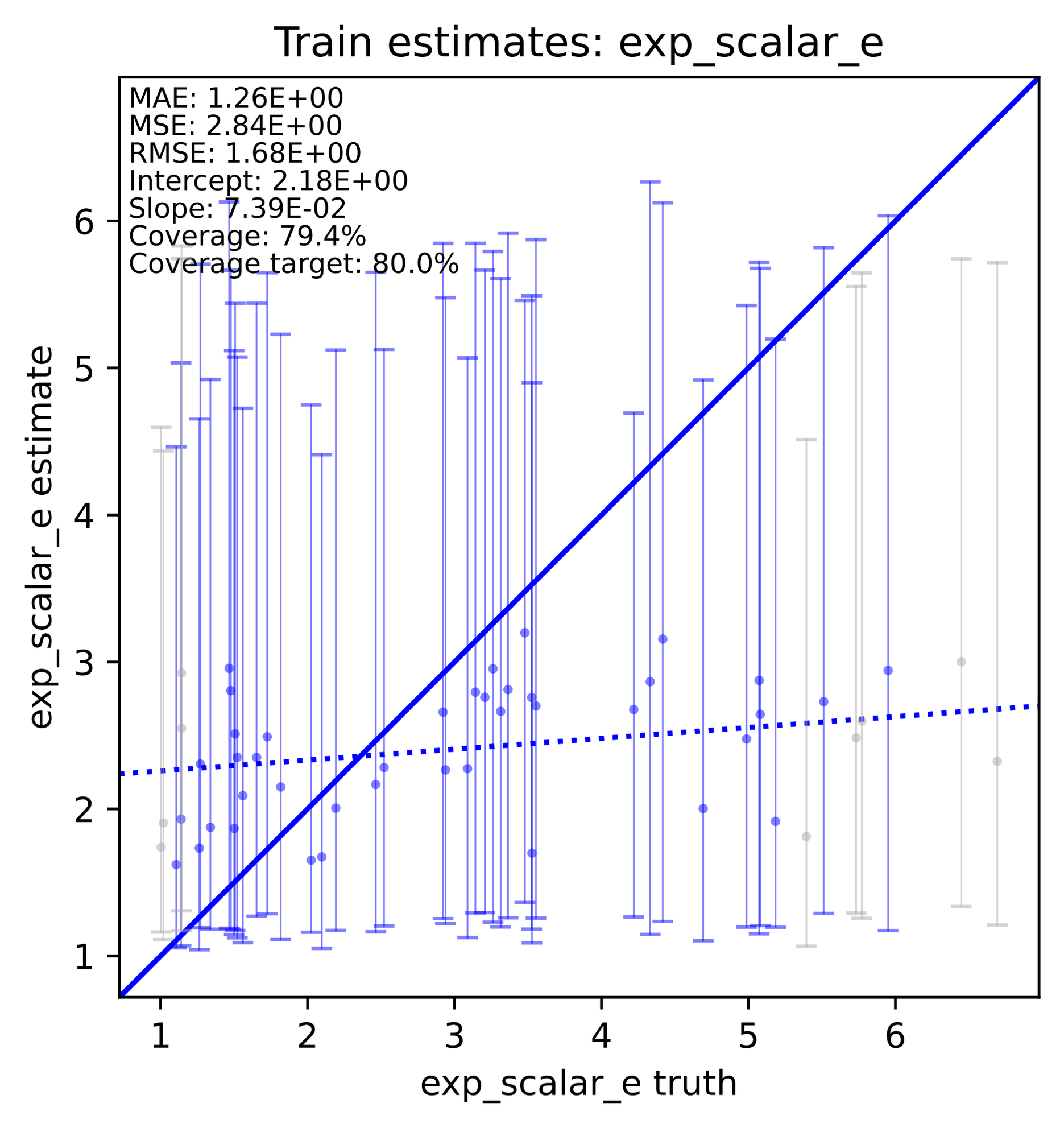

PCA: training labels
PC2 (15%)
% variance
PC
PC3 (14%)
PC4 (14%)
PC1 (31%)
PC2 (15%)
PC3 (14%)

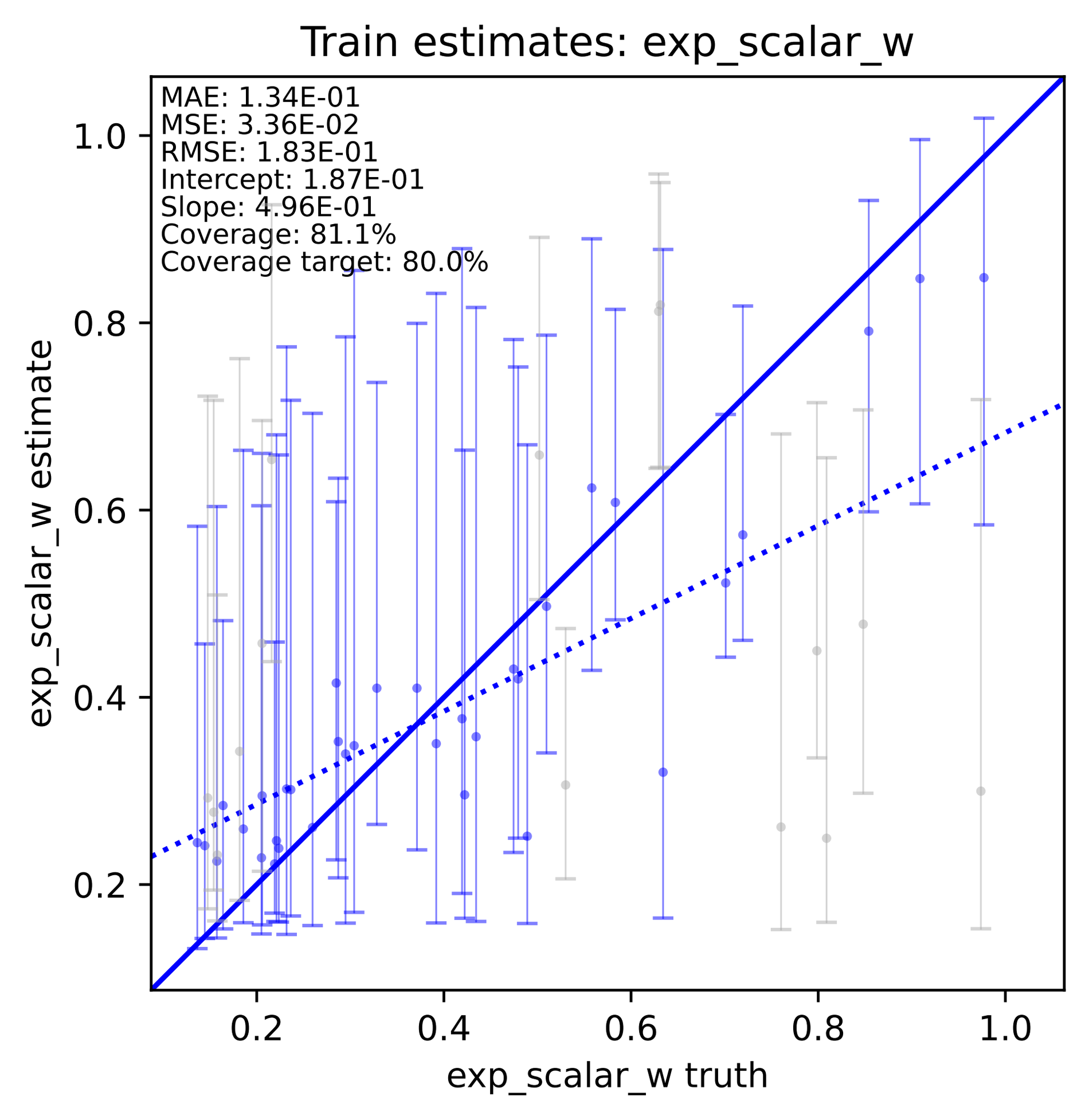

PCA: training phy. data
PC2 (7%)
% variance
PC
PC3 (2%)
PC4 (1%)
PC1 (30%)
PC2 (7%)
PC3 (2%)

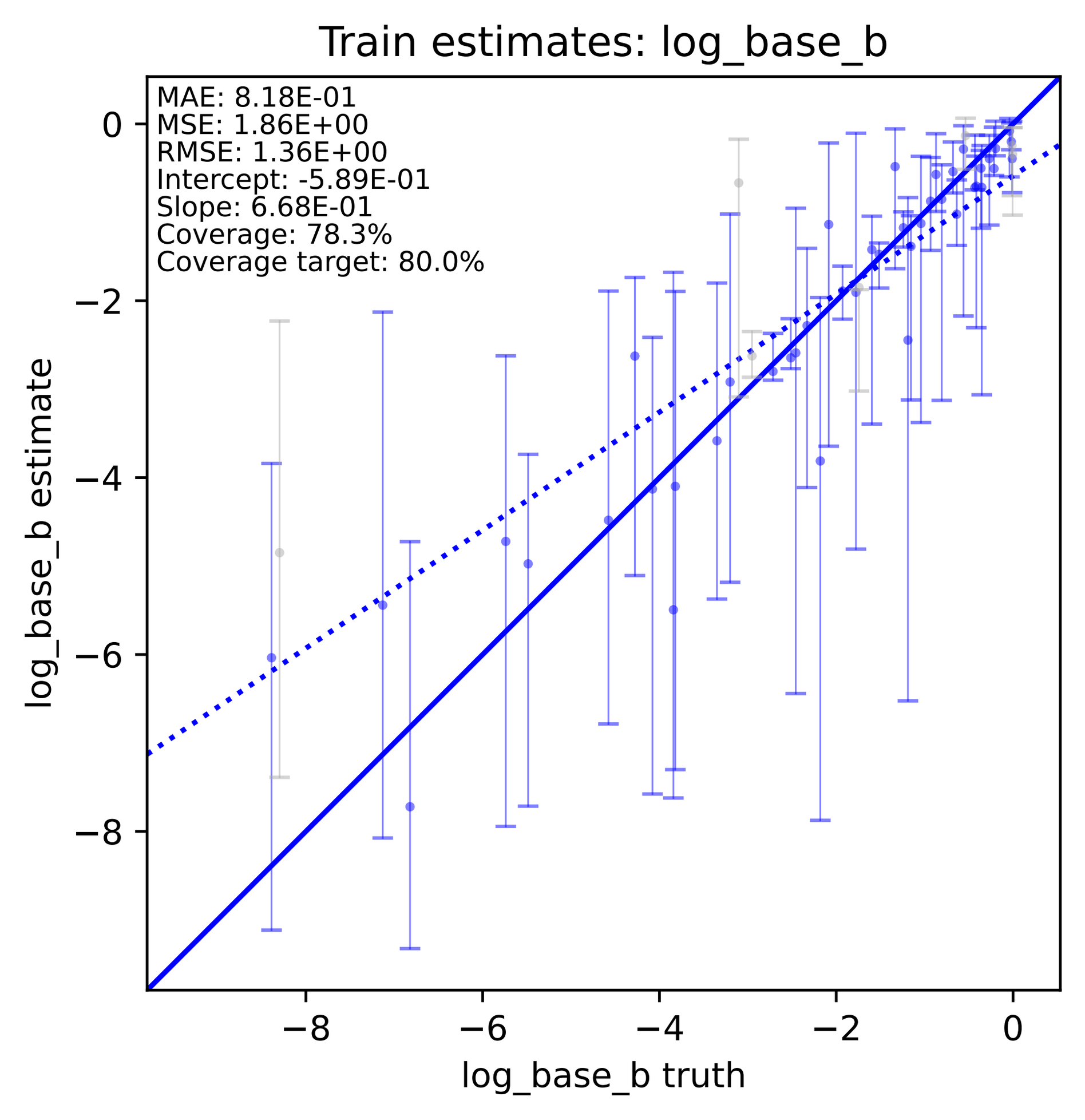

Density: training aux. data
N_bar
sim: 0.91 (0.57, 1.04)
emp: 0.99 (0.99, 0.99)
age_var
sim: 1.89 (-0.07, 2.87)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 1.48 (-0.62, 2.57)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.16 (-1.62, -0.36)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 5.67 (1.01, 322.81)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.99 (1.03, 81.32)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.77 (0.71, 2.19)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.84 (0.00, 1.58)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.61 (-0.28, 1.33)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.57 (0.61, 1.97)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.76 (1.53, 3.64)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.89 (3.66, 9.56)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 6.01 (3.04, 8.19)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 108.00 (10.00, 283.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.46, -0.07)
emp: -0.44 (-0.44, -0.44)
Density
Variable

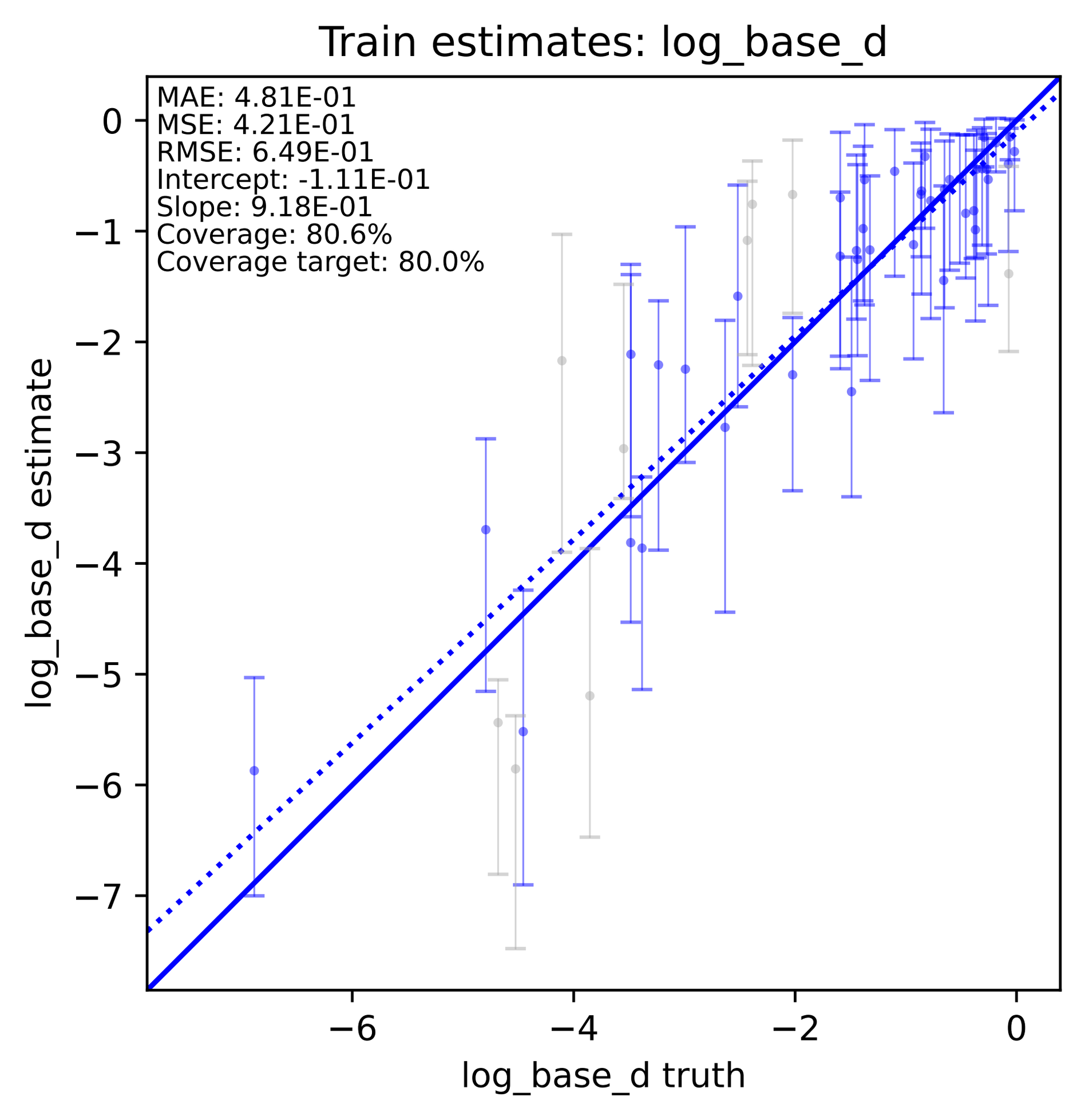

Density: training labels
exp_scalar_d_j
sim: 0.40 (0.14, 0.96)
emp: 0.29 (0.29, 0.29)
exp_scalar_e
sim: 2.28 (1.04, 6.88)
emp: 2.23 (2.23, 2.23)
exp_scalar_w
sim: 0.40 (0.14, 0.96)
emp: 0.21 (0.21, 0.21)
log_base_b
sim: -1.44 (-8.57, -0.04)
emp: -2.47 (-2.47, -2.47)
log_base_d
sim: -1.10 (-8.36, -0.03)
emp: -7.01 (-7.01, -7.01)
log_base_e
sim: -3.16 (-8.89, -0.23)
emp: -8.11 (-8.11, -8.11)
log_base_w
sim: -1.10 (-8.03, -0.03)
emp: -1.30 (-1.30, -1.30)
Density
Variable

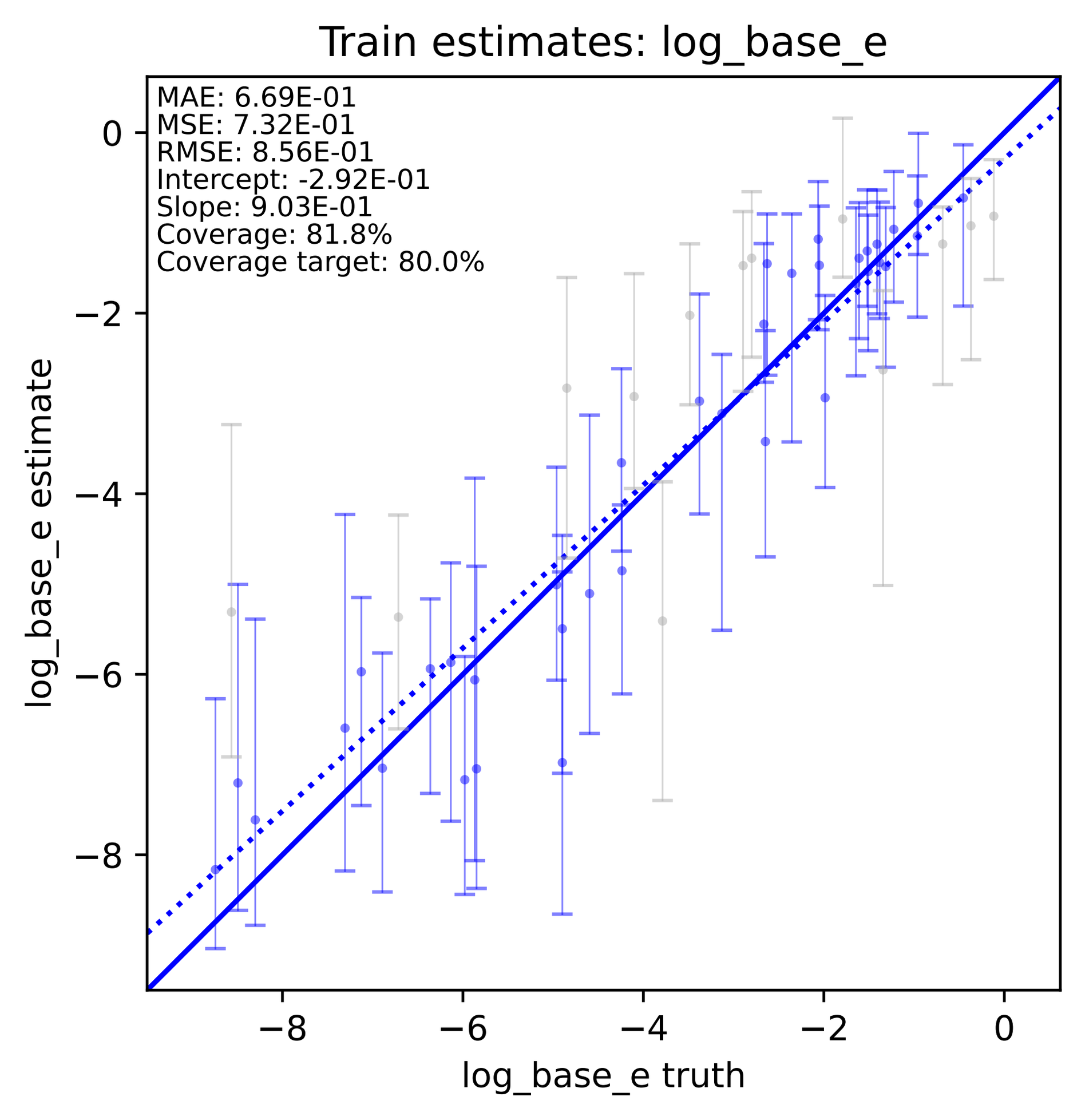

Train estimates: exp_scalar_d_j
MAE: 1.45E-01
MSE: 3.96E-02
RMSE: 1.99E-01
Intercept: 2.14E-01
Slope: 4.44E-01
Coverage: 79.2%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
1.0
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
1.0

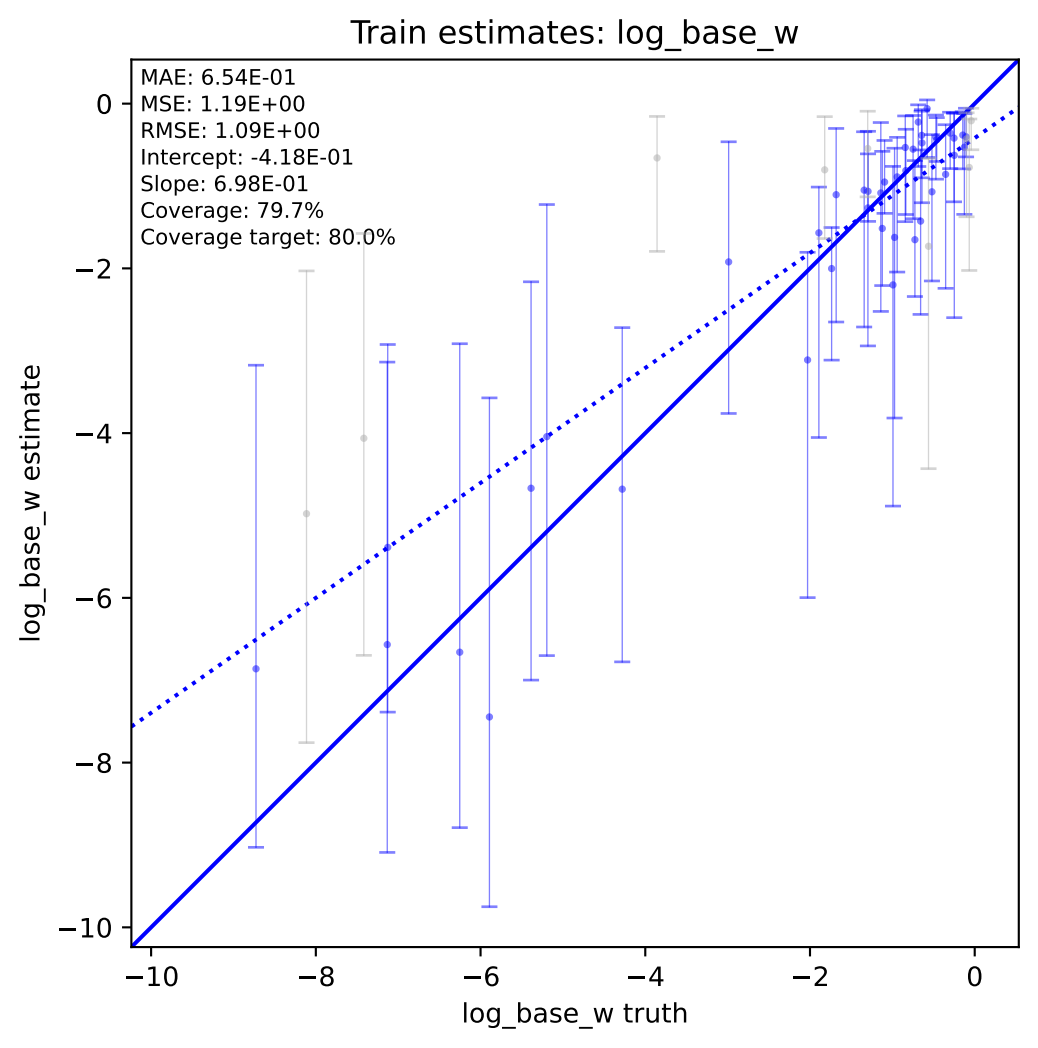

Train estimates: exp_scalar_e
MAE: 1.26E+00
MSE: 2.84E+00
RMSE: 1.68E+00
Intercept: 2.18E+00
Slope: 7.39E-02
Coverage: 79.4%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6

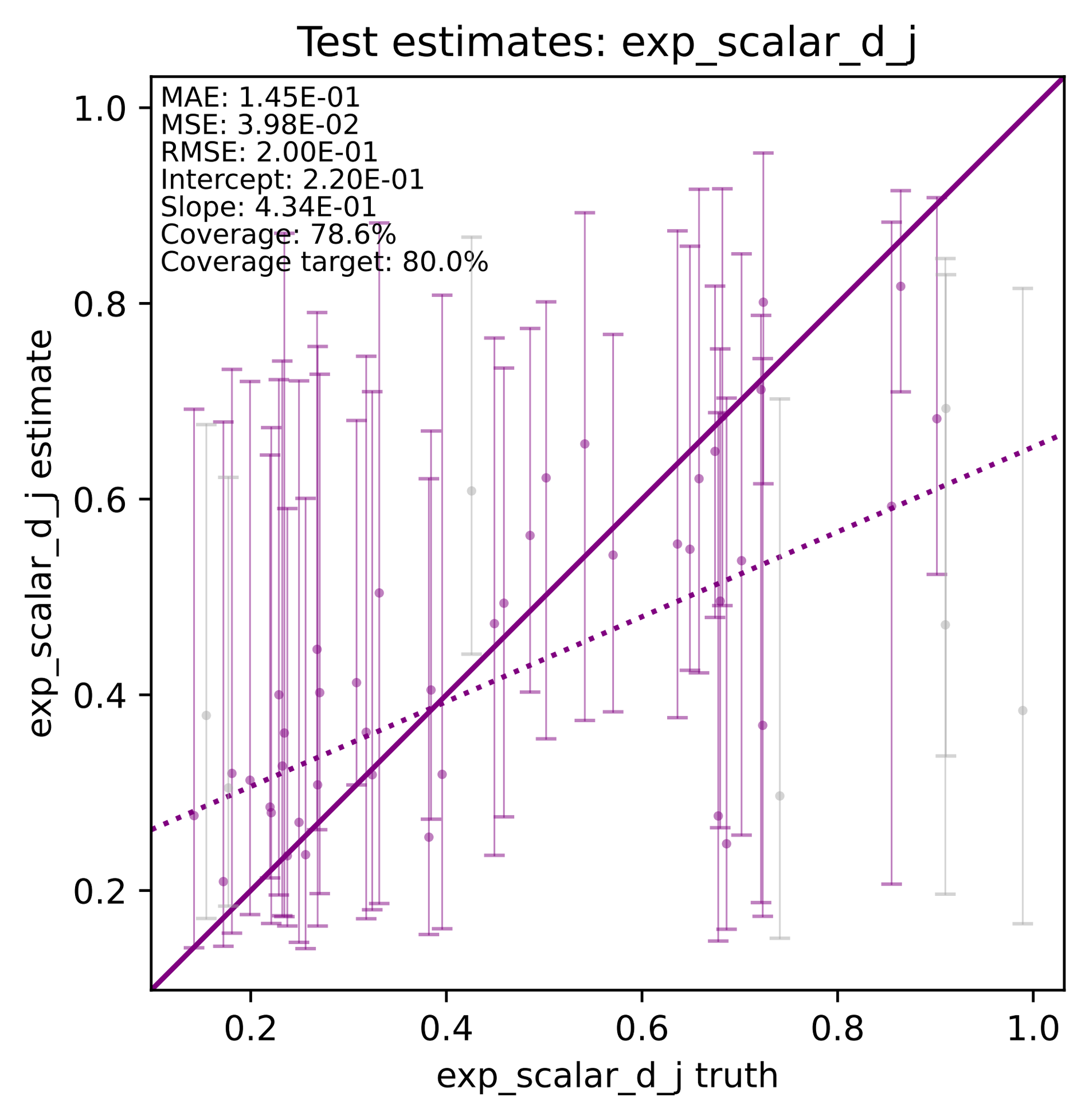

Train estimates: exp_scalar_w
MAE: 1.34E-01
MSE: 3.36E-02
RMSE: 1.83E-01
Intercept: 1.87E-01
Slope: 4.96E-01
Coverage: 81.1%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

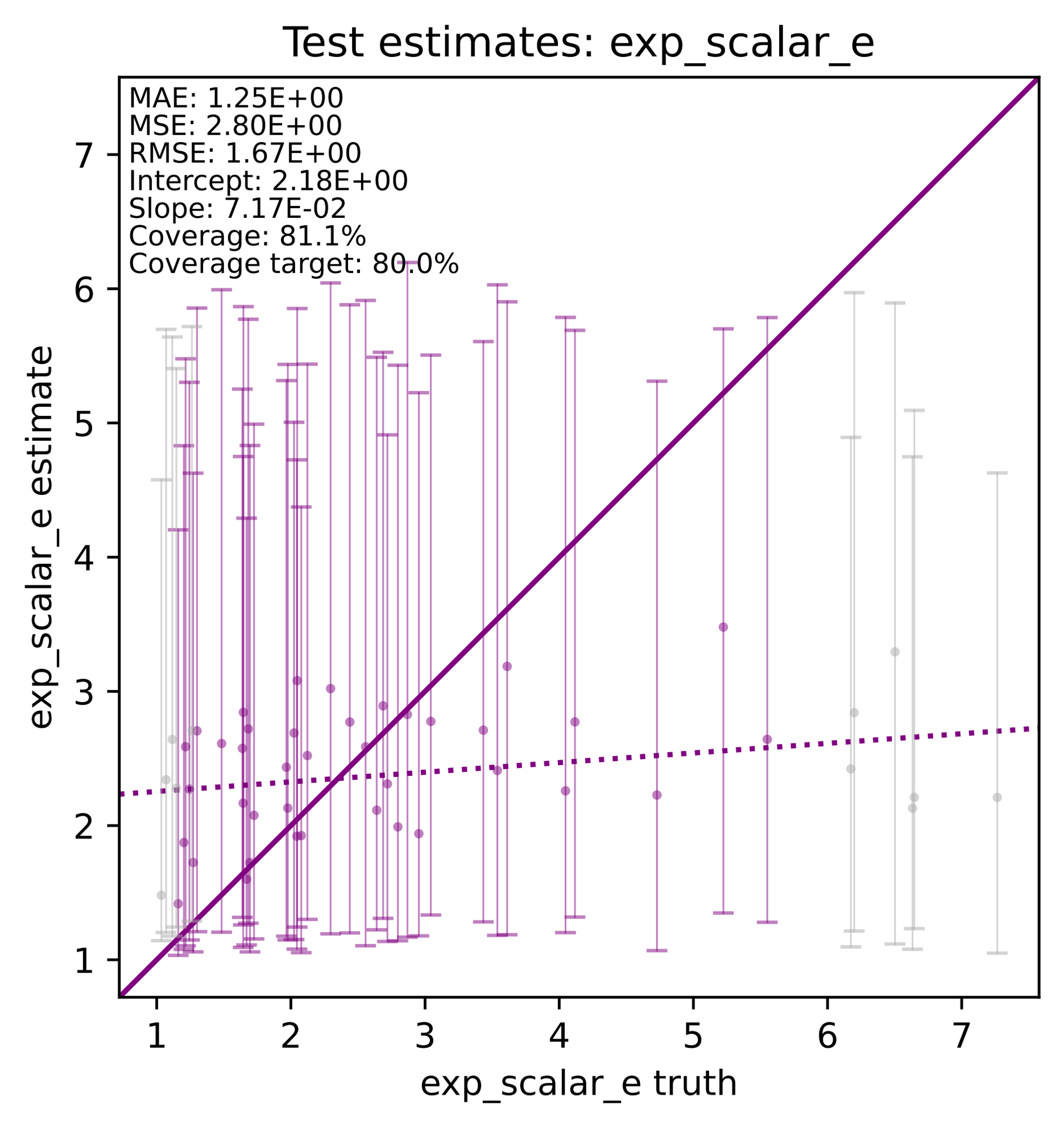

Train estimates: log_base_b
MAE: 8.18E-01
MSE: 1.86E+00
RMSE: 1.36E+00
Intercept: -5.89E-01
Slope: 6.68E-01
Coverage: 78.3%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

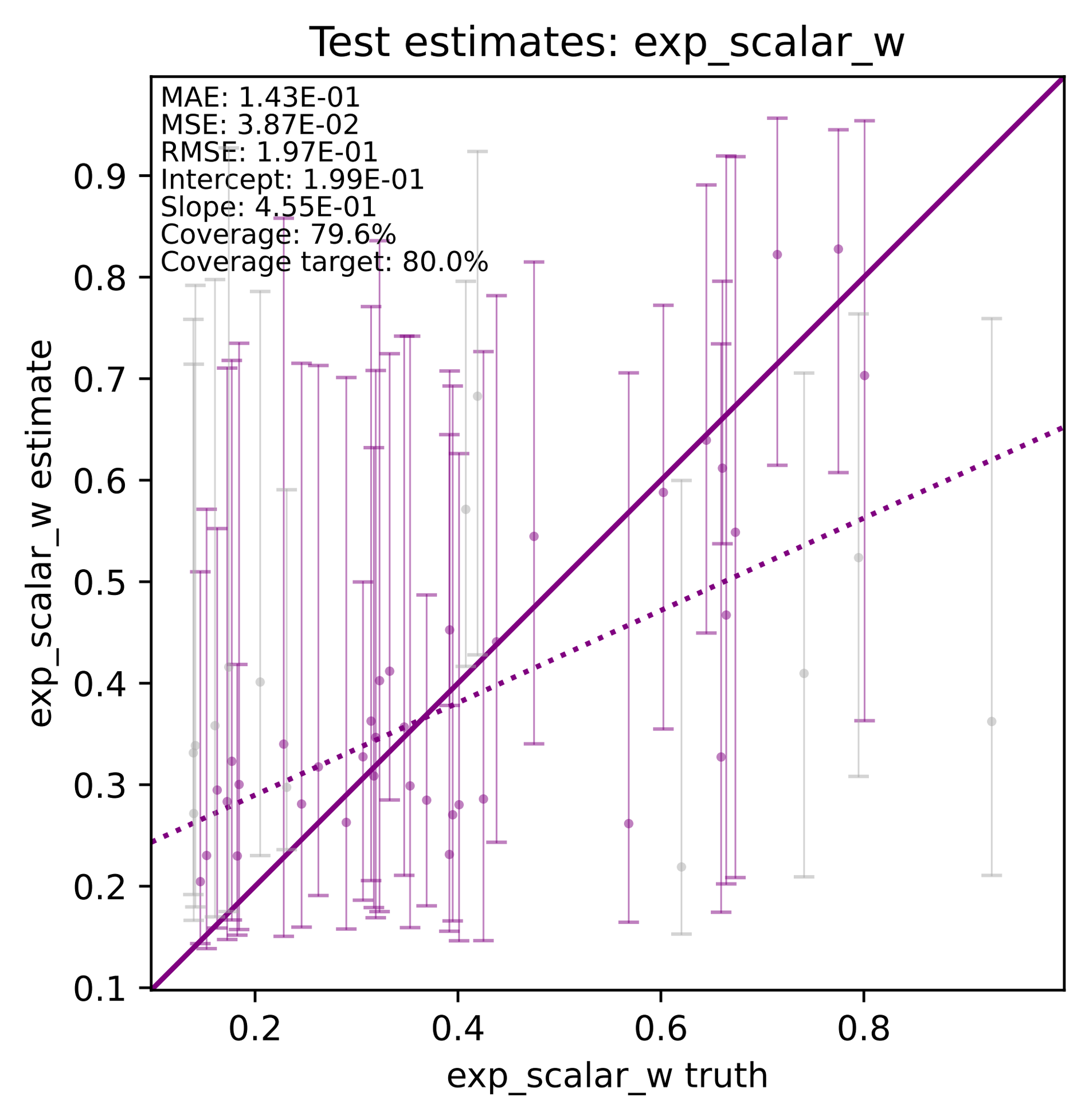

Train estimates: log_base_d
MAE: 4.81E-01
MSE: 4.21E-01
RMSE: 6.49E-01
Intercept: -1.11E-01
Slope: 9.18E-01
Coverage: 80.6%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−1
−2
−3
−4
−5
−6
−7
−6
−4
−2
0

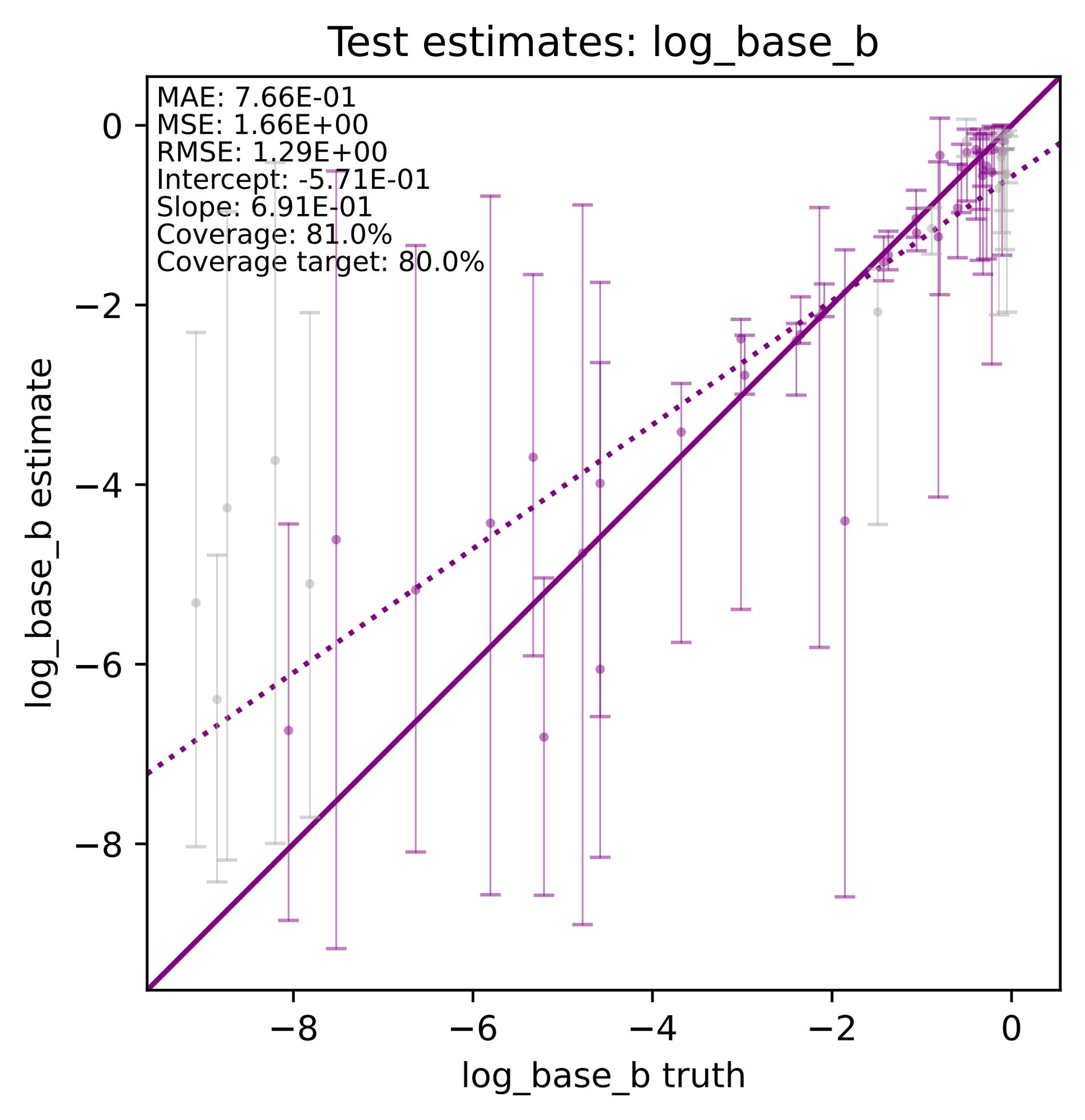

Train estimates: log_base_e
MAE: 6.69E-01
MSE: 7.32E-01
RMSE: 8.56E-01
Intercept: -2.92E-01
Slope: 9.03E-01
Coverage: 81.8%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

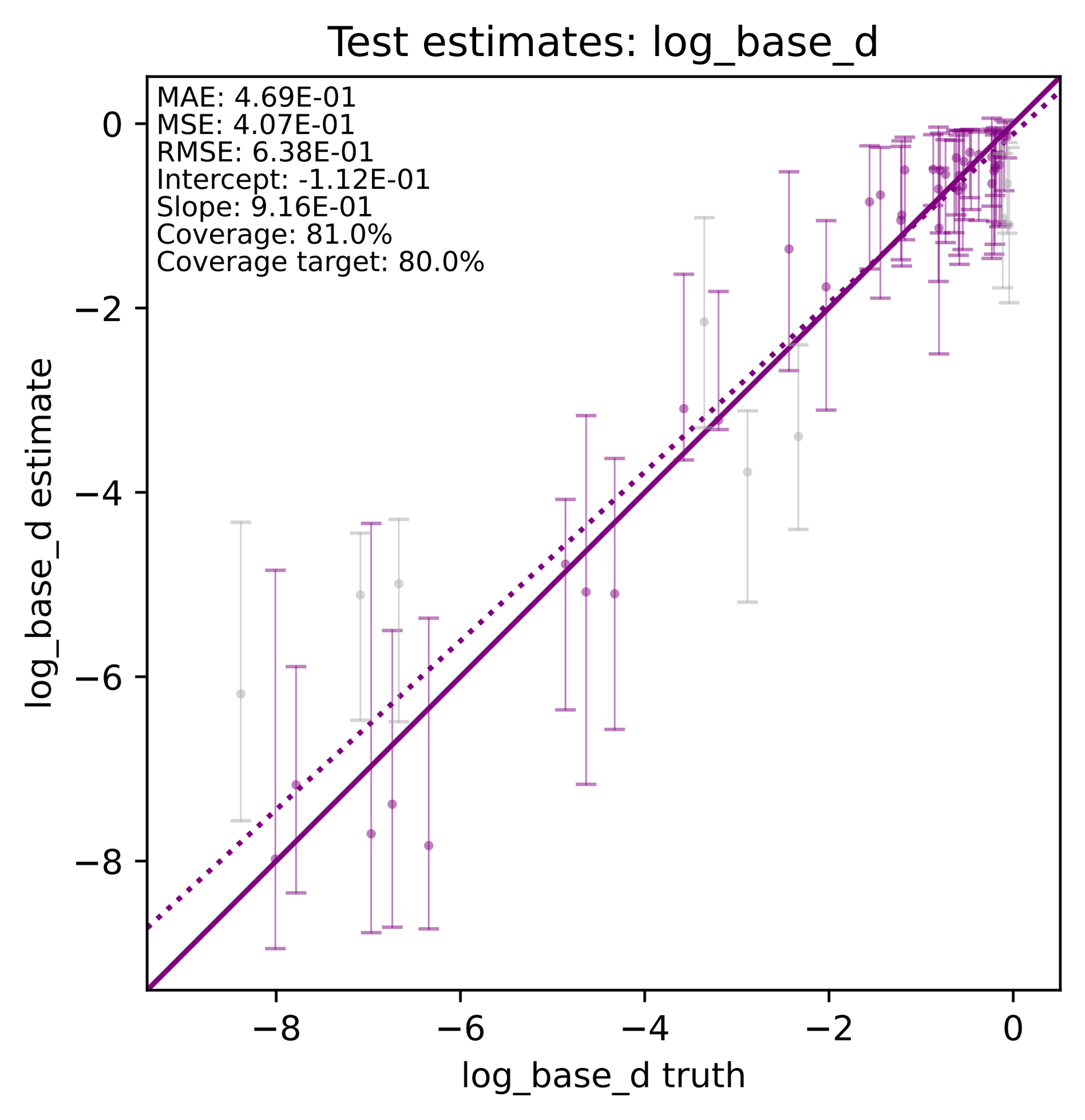

Train estimates: log_base_w
MAE: 6.54E-01
MSE: 1.19E+00
RMSE: 1.09E+00
Intercept: -4.18E-01
Slope: 6.98E-01
Coverage: 79.7%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

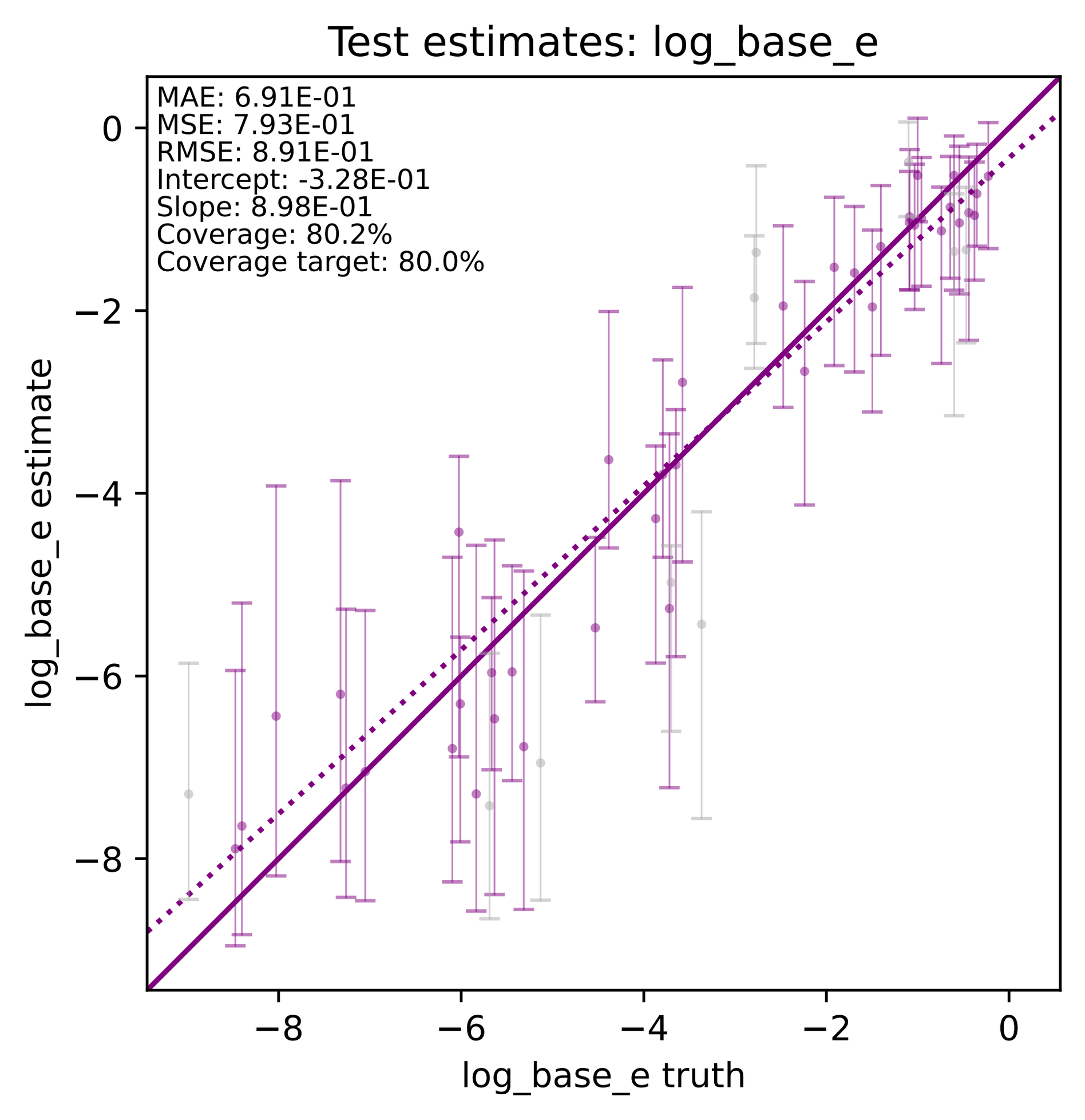

Test estimates: exp_scalar_d_j
MAE: 1.45E-01
MSE: 3.98E-02
RMSE: 2.00E-01
Intercept: 2.20E-01
Slope: 4.34E-01
Coverage: 78.6%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

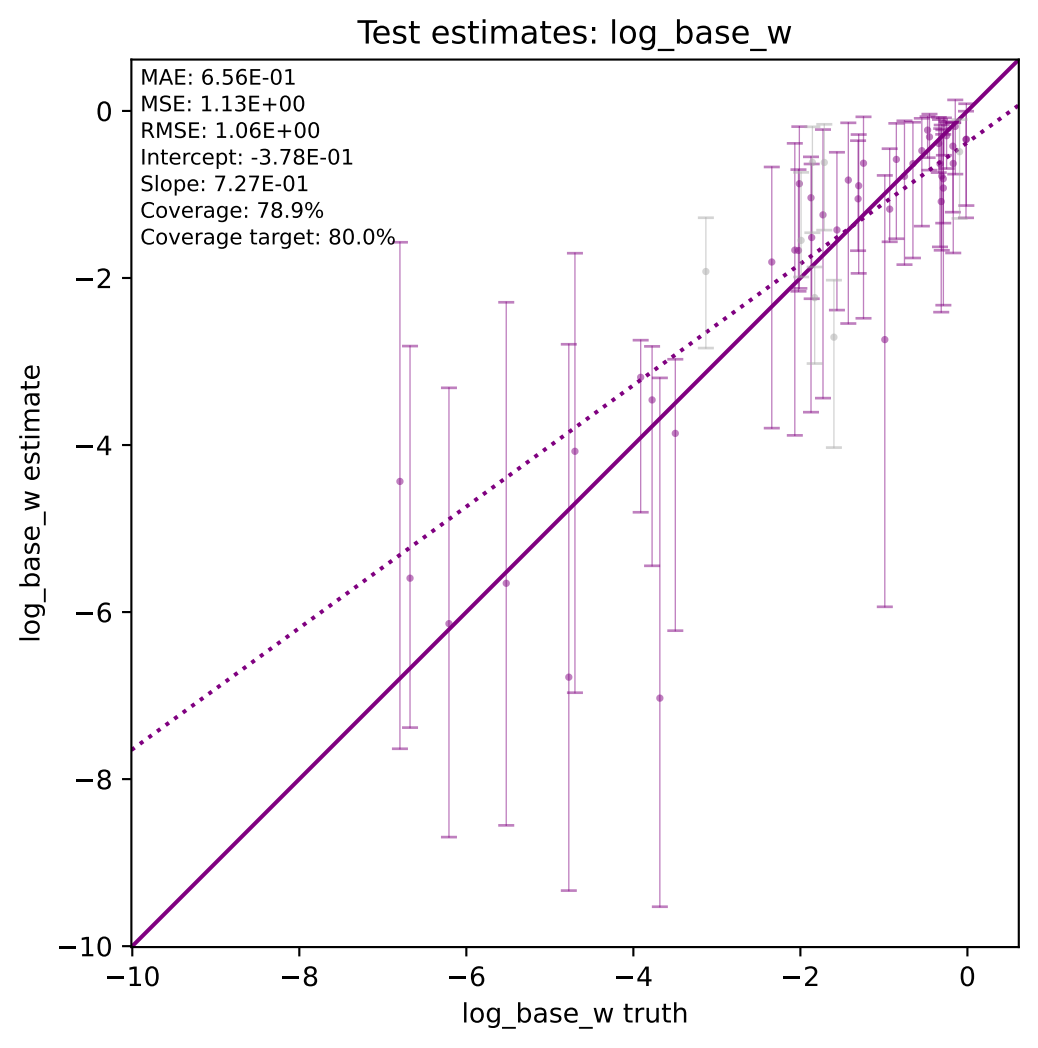

Test estimates: exp_scalar_e
MAE: 1.25E+00
MSE: 2.80E+00
RMSE: 1.67E+00
Intercept: 2.18E+00
Slope: 7.17E-02
Coverage: 81.1%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

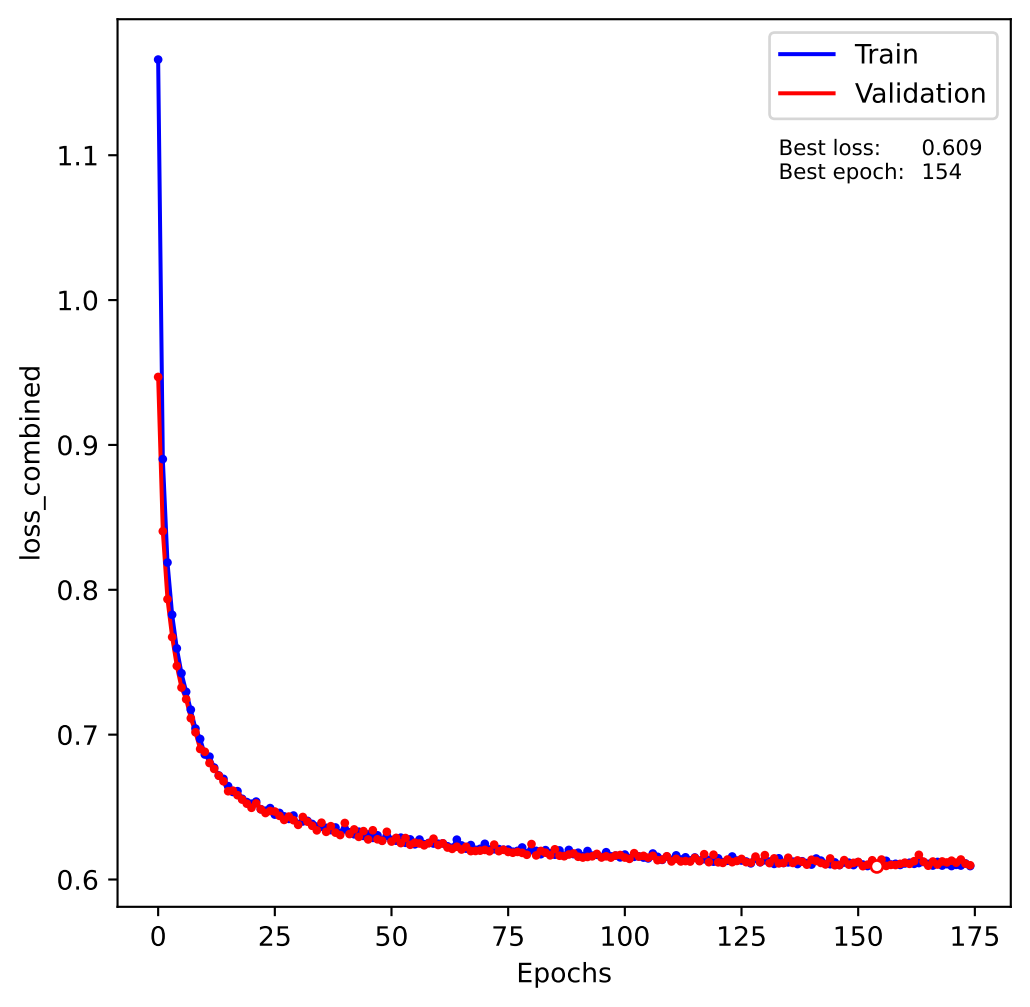

Test estimates: exp_scalar_w
MAE: 1.43E-01
MSE: 3.87E-02
RMSE: 1.97E-01
Intercept: 1.99E-01
Slope: 4.55E-01
Coverage: 79.6%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
0.2
0.4
0.6
0.8

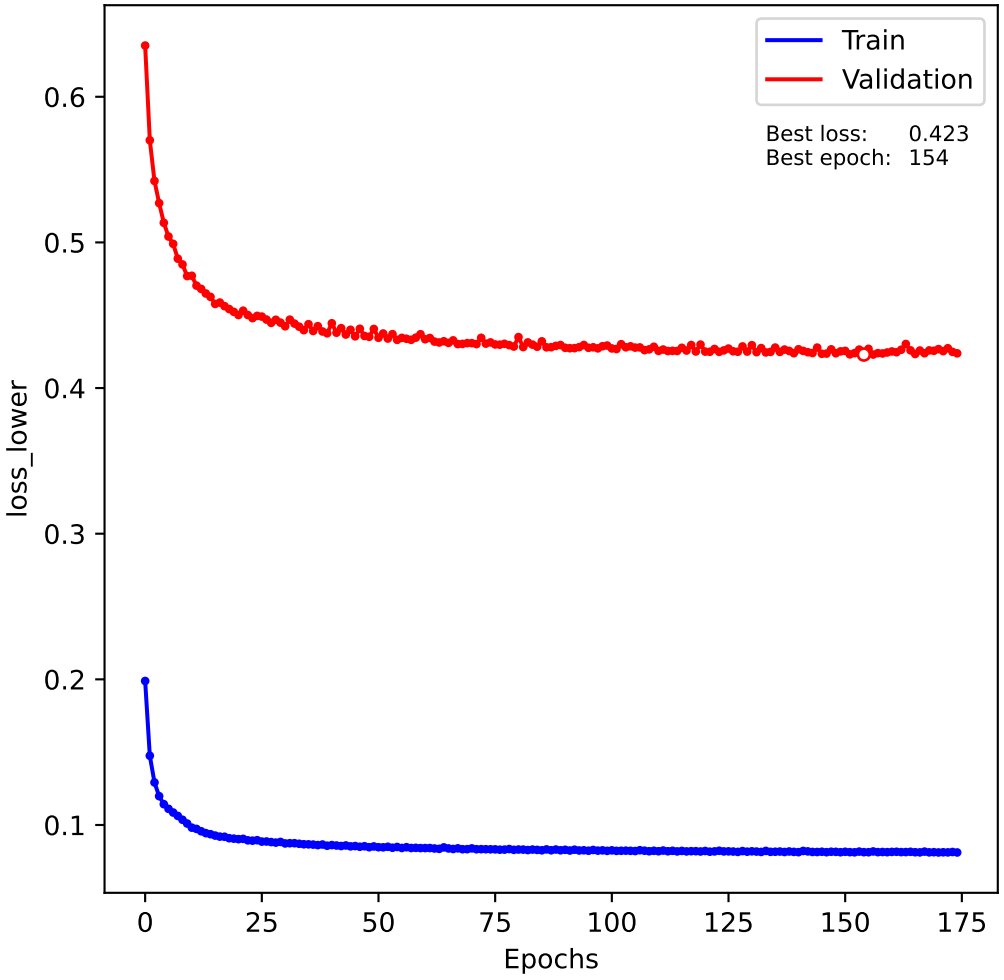

Test estimates: log_base_b
MAE: 7.66E-01
MSE: 1.66E+00
RMSE: 1.29E+00
Intercept: -5.71E-01
Slope: 6.91E-01
Coverage: 81.0%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

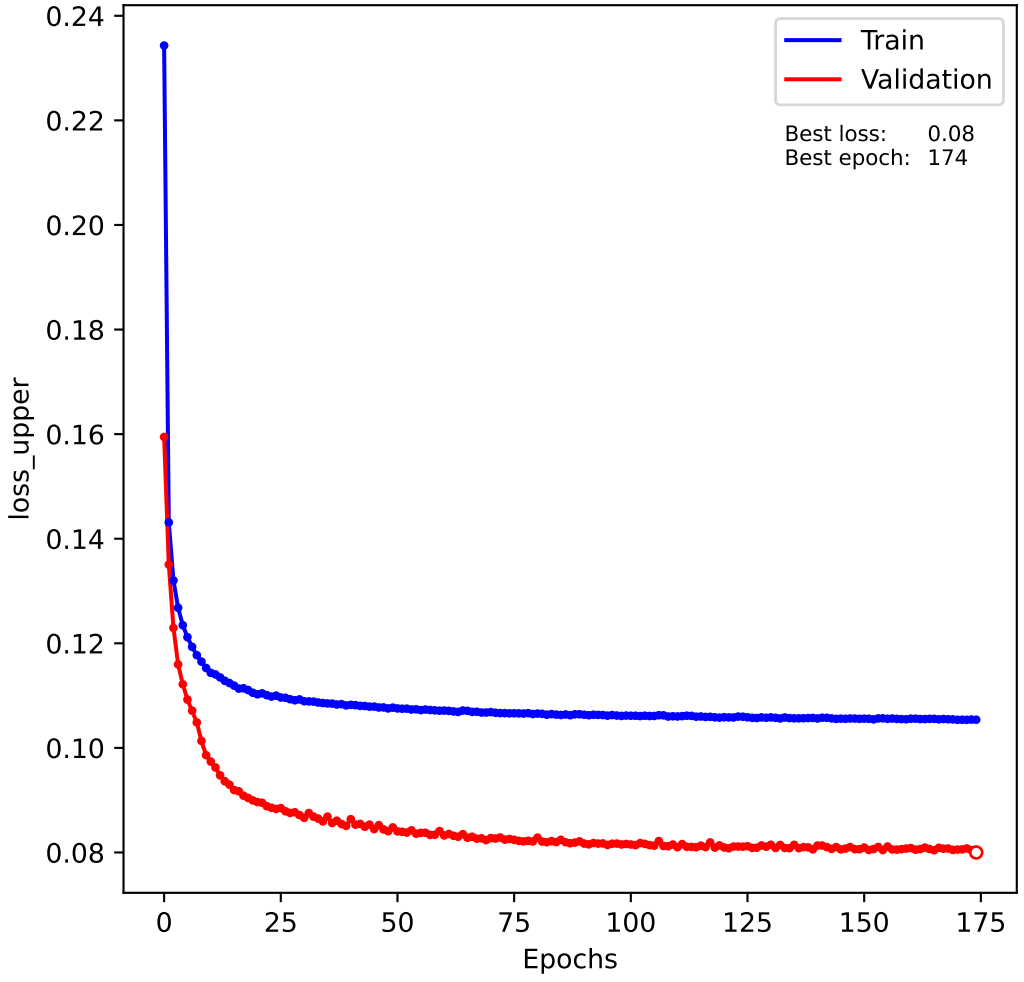

Test estimates: log_base_d
MAE: 4.69E-01
MSE: 4.07E-01
RMSE: 6.38E-01
Intercept: -1.12E-01
Slope: 9.16E-01
Coverage: 81.0%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

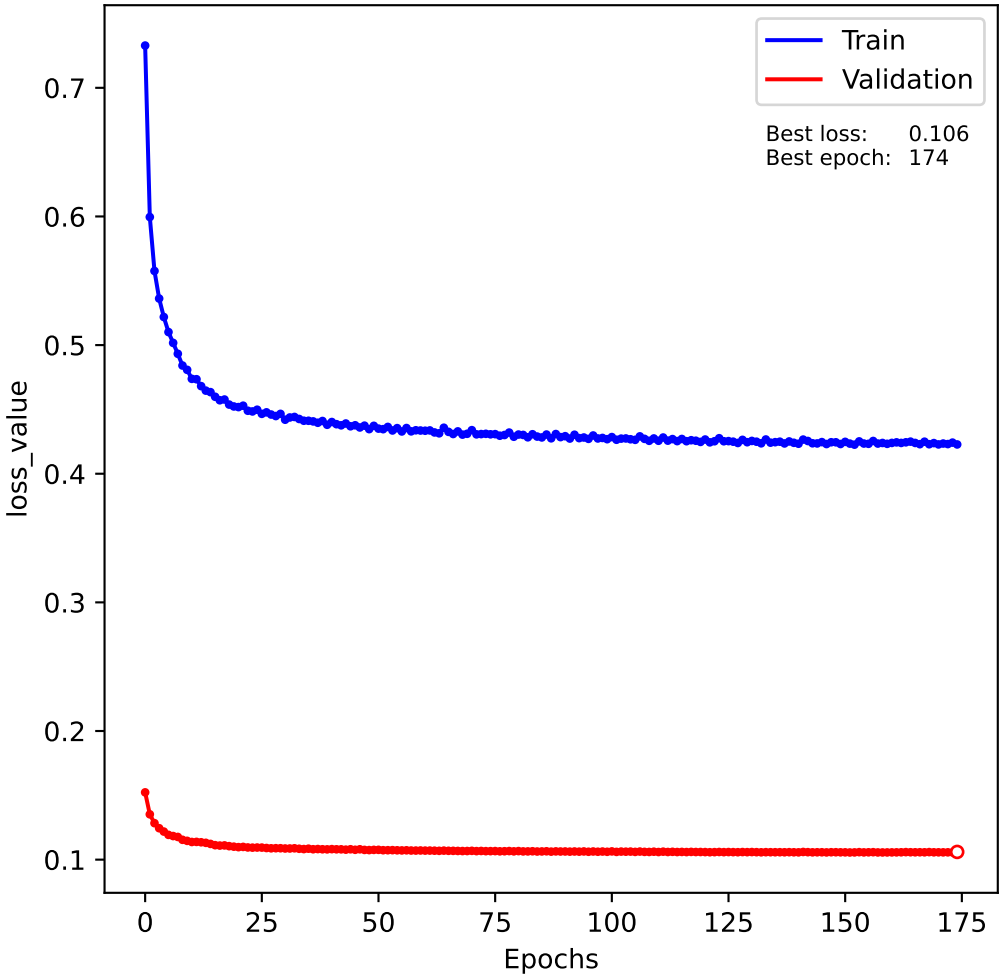

Test estimates: log_base_e
MAE: 6.91E-01
MSE: 7.93E-01
RMSE: 8.91E-01
Intercept: -3.28E-01
Slope: 8.98E-01
Coverage: 80.2%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

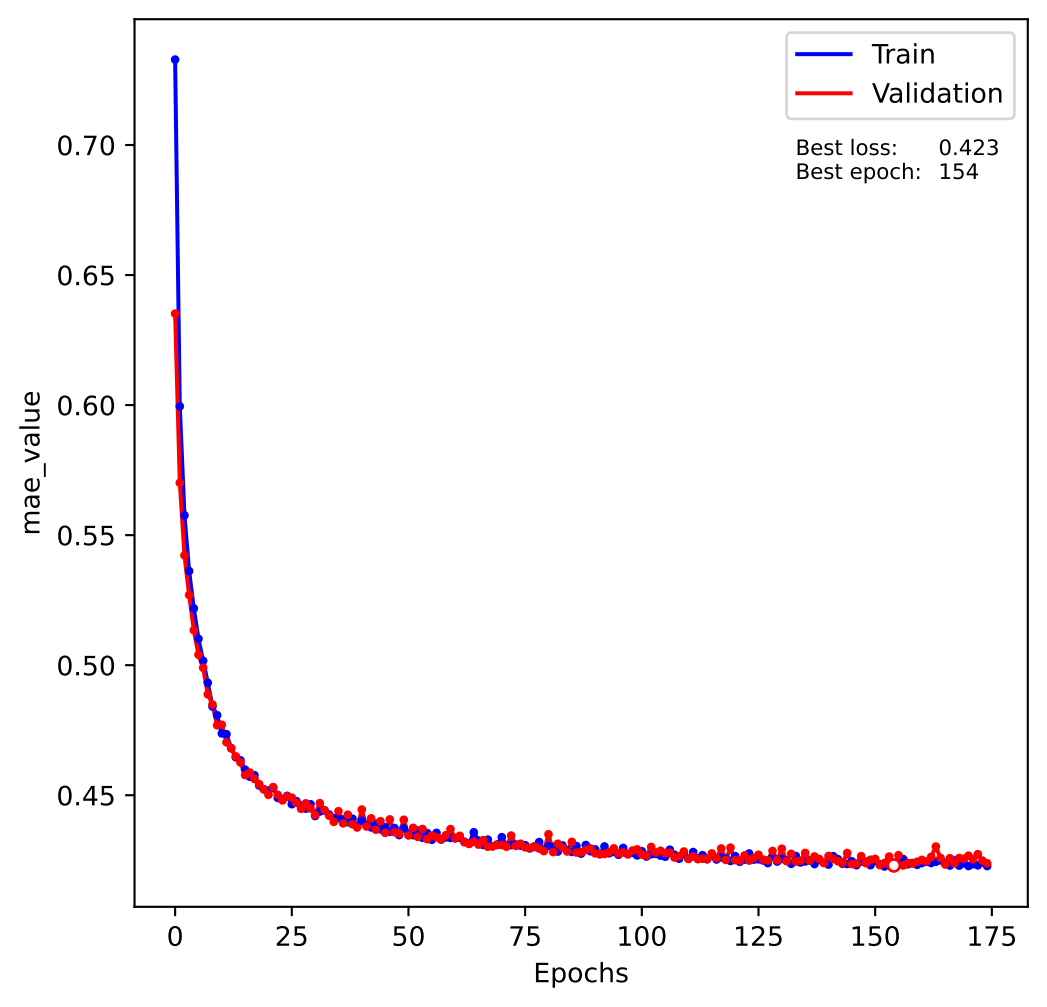

Test estimates: log_base_w
MAE: 6.56E-01
MSE: 1.13E+00
RMSE: 1.06E+00
Intercept: -3.78E-01
Slope: 7.27E-01
Coverage: 78.9%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

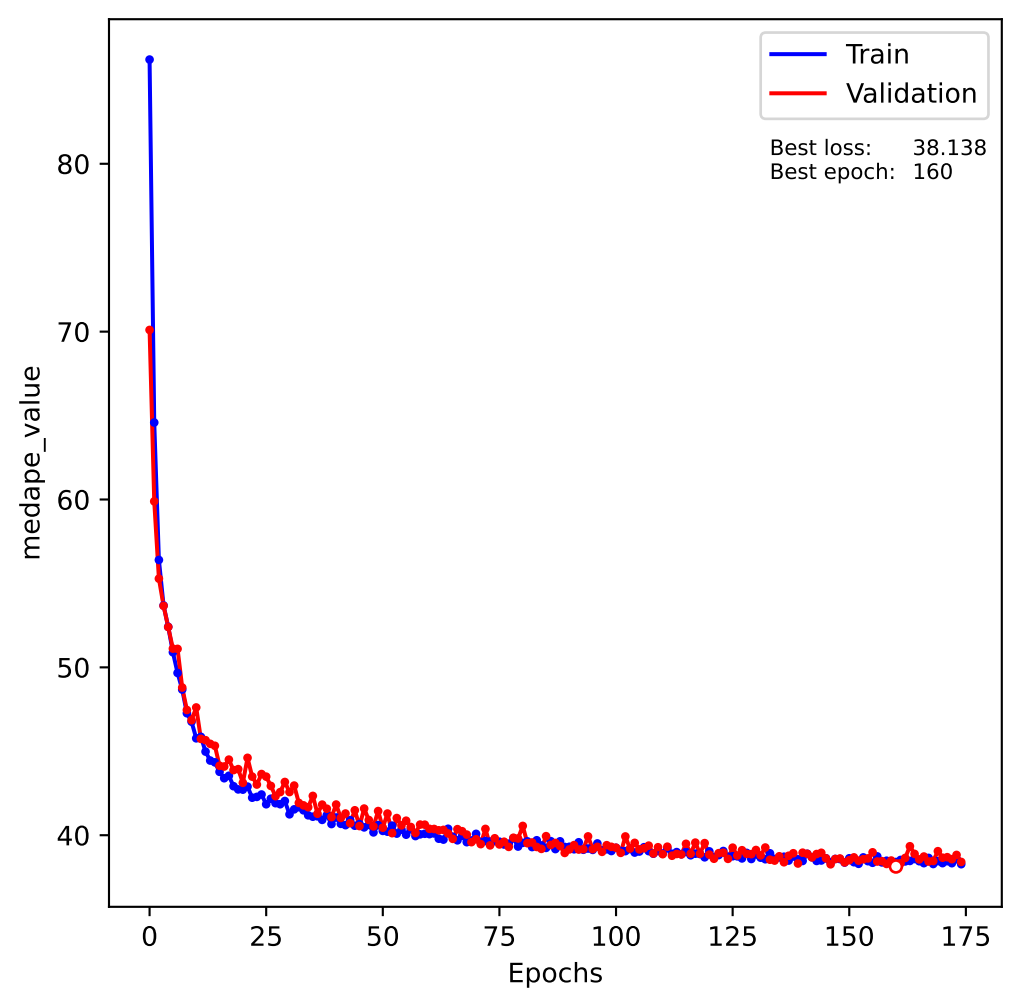

Train
Validation
Best loss: 0.609
Best epoch: 154
loss_combined
1.1
1.0
0.9
0.8
0.7
0.6
0
25
50
75
100
125
150
175
Epochs

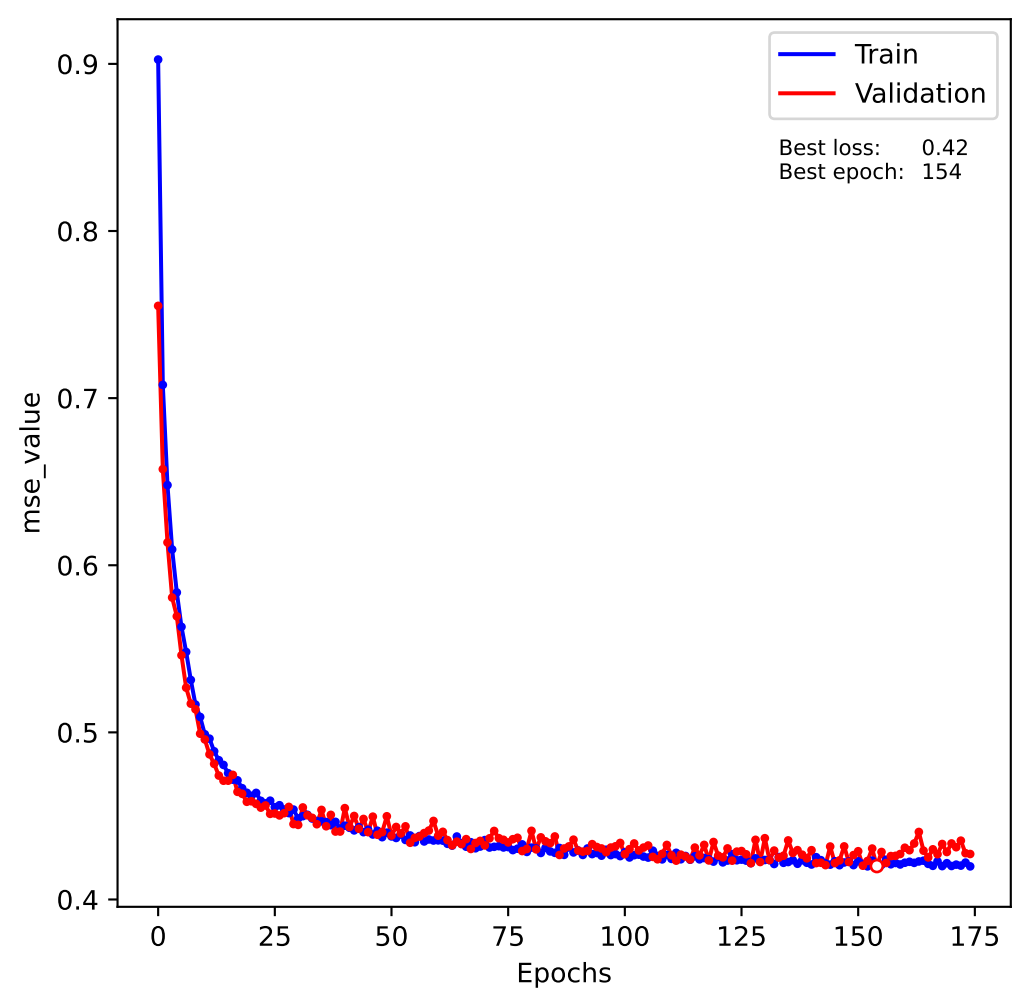

Train
Validation
Best loss: 0.423
Best epoch: 154
loss_lower
0.6
0.5
0.4
0.3
0.2
0.1
0
25
50
75
100
125
150
175
Epochs

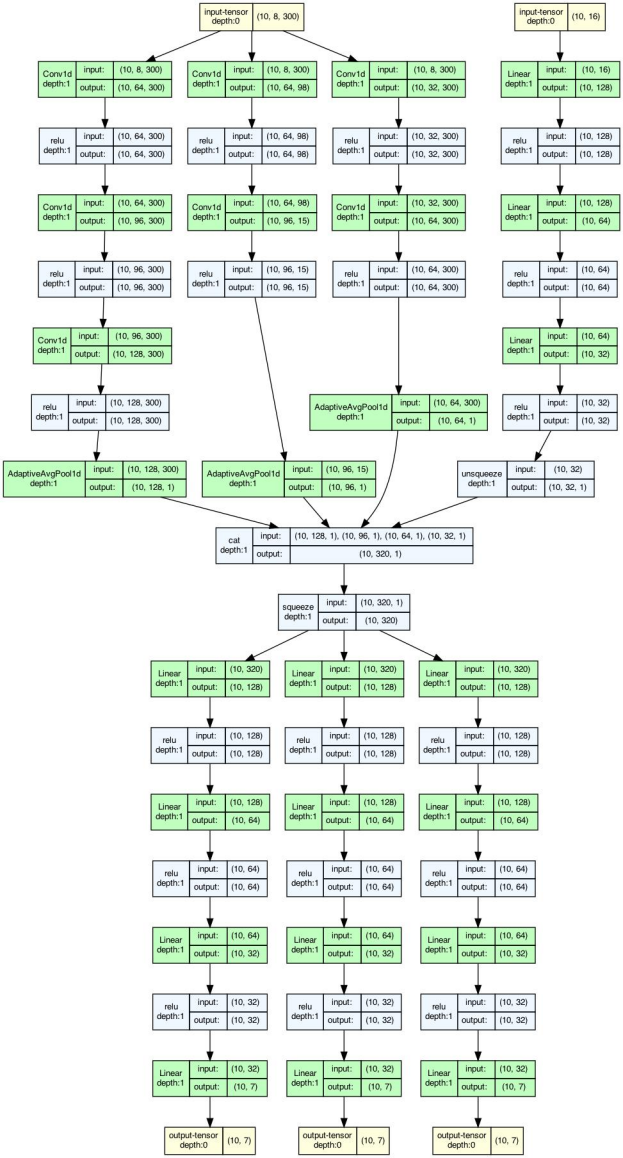

Train
Validation
Best loss: 0.08
Best epoch: 174
loss_upper
Epochs
0.24
0.22
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0
25
50
75
100
125
150
175


Train
Validation
Best loss: 0.106
Best epoch: 174
loss_value
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
25
50
75
100
125
150
175
Epochs


Train
Validation
Best loss: 0.423
Best epoch: 154
0.70
0.65
0.60
0.55
0.50
0.45
mae_value
0
25
50
75
100
125
150
175
Epochs


Train
Validation
Best loss: 38.138
Best epoch: 160
medape_value
Epochs
80
70
60
50
40
0
25
50
75
100
125
150
175


Train
Validation
Best loss: 0.42
Best epoch: 154
mse_value
Epochs
0.9
0.8
0.7
0.6
0.5
0.4
0
25
50
75
100
125
150
175


input-tensor depth:0 (10, 8, 300)
input-tensor depth:0 (10, 16)
Conv1d depth:1 input: (10, 8, 300) output: (10, 64, 300)
Conv1d depth:1 input: (10, 8, 300) output: (10, 64, 98)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Linear depth:1 input: (10, 16) output: (10, 128)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64, 98) output: (10, 64, 98)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 128) output: (10, 128)
Conv1d depth:1 input: (10, 64, 300) output: (10, 96, 300)
Conv1d depth:1 input: (10, 64, 98) output: (10, 96, 15)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 96, 300) output: (10, 96, 300)
relu depth:1 input: (10, 96, 15) output: (10, 96, 15)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 96, 300) output: (10, 128, 300)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 128, 300) output: (10, 128, 300)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
relu depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 128, 300) output: (10, 128, 1)
AdaptiveAvgPool1d depth:1 input: (10, 96, 15) output: (10, 96, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1) output: (10, 320, 1)
squeeze depth:1 input: (10, 320, 1) output: (10, 320)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 7)
Linear depth:1 input: (10, 32) output: (10, 7)
Linear depth:1 input: (10, 32) output: (10, 7)
output-tensor depth:0 (10, 7)
output-tensor depth:0 (10, 7)
output-tensor depth:0 (10, 7)